ImpactECON Labor Database: Technical Documentation

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1 Introduction

This paper documents the construction of the ImpactECON labor database. It explains what the database contains, how it was built, and what to keep in mind when interpreting or citing it.

The ImpactECON Labor database is a harmonized, cross-country labor database covering labor force, and employment broken down by education level, occupation, and economic activity/sector. The database is built to be compatible with the GTAP (Global Trade Analysis Project) database.[1] The current version is compatible with the GTAP 12 Database and is consistent with the year 2023.

While labor quantities are not explicitly included in the standard GTAP model, their inclusion could assist in modelling several issues, such as the distinction between the labor supply and employment or unemployment,[2] the movement of labor between activities or even occupations, and the dynamics of labor supply by education and demand by occupation.

The underlying data for the ImpactECON labor data come from several sources:

- Population and total labor force by country: World Bank, by country and year, from 2000 onward.
- Labor force and unemployment by education level and country: ILO (ILOSTAT), reported under three alternate education classifications (aggregate levels, ISCED-11, ISCED-97).
- Employment by education and occupation by country: ILO (ILOSTAT), reported under two alternative occupation classifications (ISCO-08 and ISCO-88)
- Employment by occupation and economic activity by country: ILO (ILOSTAT), reported under the two alternate occupation classifications (ISCO-08 and ISCO-88) and up to four alternative activity classifications.
- Wage bill by occupation, activity and region, used only to fill and cross-check occupation/activity detail where a country has no direct survey data: GTAP 12 Database (value of factor use by occupation, activity and region, header “VFM”).

The ImpactECON labor database is compatible with the GTAP database, covering all GTAP regions, occupations and activities, for a single target year. Not every country (in the World Bank and ILO datasets) reports every breakdown for every year. Hence data gaps must be estimated or ‘filled’. The following principles are followed during the filling process:

- Real reported data are always preferred to an estimate.[3]
- More detailed classifications are preferred to less detailed: less detailed data are only used to fill when the more detailed data is unavailable.
- Nearest-year matching: if a country's data isn't available for the target year, the closest year that is available is used and rescaled to the target year using the ratio of population or labor force between the two years.

[1] Aguiar, A., Baldos, U., Chepeliev, M., Corong, E., & Simonato, T. (2025). The Global Trade Analysis Project (GTAP) Data Base: Version 12. *Journal of Global Economic Analysis*, 10(2), 1-45. Retrieved from https://doi.org/10.21642/JGEA.100201AF

[2] Walmsley, T. L., and A. Strutt (2011). “Implications of the Global Financial Crisis for China: A Dynamic CGE Analysis to 2020”, Economics Research International, October, 9 pages. https://doi.org/10.1155/2011/926484

[3] The exception to this is unemployment. Where unemployment was found to be greater than 60 percent of the labor force, the unemployment rate is ignored and a regional average used instead. Hence, extreme reported unemployment rates will not appear in the final output even if ILO's raw data contained them. Such high unemployment rates are likely the result of errors or extreme, temporary, situations which are unlikely to be reflected in the GTAP database.

- Remainder / consistency filling: where a country reports totals and most — but not all — of a breakdown (e.g. 4 of 5 ILO education categories), the missing piece is filled as the gap between the reported total and the reported components, rather than left blank.
- Regional-average fallback: if a country has no usable breakdown at all for a given series, it is filled using the average rate or share observed across other countries in its region (and, if the region itself has no data, the world average). This is the least precise method and is used only as a last resort.
- GTAP-anchored filling for occupation/activity detail: where a country has no ILO occupation or activity breakdown, employment by activity is instead obtained from GTAP's own wage-bill data — so for these countries, employment figures are implied from the GTAP data rather than directly observed.

The 'filling' method has a number of limitations, but most notably a meaningful share of the occupation by activity detail, and some of the education-level detail, for smaller or data-sparse countries is estimated rather than directly reported.

2 The ImpactECON Labor Database Structure

Table 1 illustrates the structure of the ImpactECON Labor database. It contains 4 headers including the labor supply (LFE) and employment (LDE) by education level, where the difference between these two arrays is unemployment; the link between the education levels and the occupations of workers (LDEO); and employment by occupation and activity (LDOA).

In this new version of the labor database, the link between education and occupation (LDEO) is based on real data obtained from the ILO. In past versions of the ImpactECON labor database, the link between education levels and occupations was obtained by assuming that that workers with higher education levels (tertiary) were employed primarily in higher skilled occupations (professionals and technicians), while workers with secondary education were primarily employed in clerical and service occupations, and the least skilled employed as other low skilled workers. Where gaps existed, these were filled with workers with the closest level of education. For instance, in countries with low education rates, the higher skilled occupations often included workers with only secondary education, and in countries with high education rates clerical and service occupations often contained tertiary educated workers. The improvement in the ILO data here represents a great improvement over previous versions of the ImpactECON labor database.

The fourth header (LDOA) containing employment by occupation and activity provides the labor quantities associated with the header VFM in the GTAP database.

Table 1: The ImpactECON Labor Database Headers

Variable	Meaning	Dimensions
LFE	Labor supply (labor force)	by education level[a] × country/region[b]
LDE	Employment (labor demand)	by education level × country/region
LDEO	Employment	by education level × occupation[c] × country/region
LDOA	Employment	by occupation × economic activity/sector[d] × country/region

a. Education levels: 3 education levels (basic, secondary and tertiary)
b. GTAP regions: 163 countries/regions (v12, February 2026)

c. GTAP occupations: 5 occupation categories (Professionals, Technicians and Associate Professionals, Clerks, Sales and Service workers, and Other Low Skilled and Agricultural workers).
d. GTAP activities/sectors: https://www.gtap.agecon.purdue.edu/databases/v12/v12_sectors.aspx
e. The final data show numbers of people, not full-time equivalents.

3 Data Sources

Table 2 contains the sources of the underlying data used to construct the ImpactECON labor database. The World Bank data portal is the most complete, containing population and labor force data for most countries and years. We therefore prioritize consistency with this data. Population data for roughly 34 very small countries/ territories are not covered by the World Bank series (e.g. Guernsey, Jersey, Falkland Islands, Taiwan). Estimates of population for these countries are obtained from an internet search. With the exception of Taiwan, these countries are small dependencies or islands which are used in GTAP's rest of regions. In the case of Taiwan, all the data is estimated based on the data of East Asia, with the exception of population, which was obtained from a web search.

Table 2: Data in ImpactECON Labor Database

Source	Series / file	Used for
World Bank	SP.POP.TOTL	Total population by country and year
World Bank	SL.TLF.TOTL.IN	Total labor force by country and year
ILOSTAT	EAP_TEAP_SEX_EDU_NB_A	Labor force by education (aggregate levels, ISCED-11, ISCED-97)(1000's)
ILOSTAT	UNE_TUNE_SEX_EDU_NB_A	Unemployment by education (aggregate levels, ISCED-11, ISCED-97) (1000's)
ILOSTAT	EMP_TEMP_SEX_OCU_EDU_NB_A	Employment by occupation (ISCO-08 and ISCO-88) and education (aggregate levels, ISCED-11, ISCED-97) (1000's)
ILOSTAT	EMP_TEMP_ECO_OCU_NB_A	Employment by occupation (ISCO-08 and ISCO-88) and economic activity (ISIC Rev 4, ISIC Rev 3.1, Aggregate, Broad sector) (1000's)
Internet search	MISSINGPOP table	2023/2024 population estimates for micro-territories and dependencies not in the World Bank series (e.g. Falkland Islands, Guernsey, Jersey, Taiwan, Western Sahara) are obtained from an internet search.

The ILO data is less consistent than the World Bank data. Data is often available for only one year and different ILO definitions are often used for education, occupation and activities depending on the country and the year in which the country last reported. Over time the ILO definitions have become more detailed and hence data reported using the newer definitions are generally preferred to those using older definitions, except in the case of education where the aggregate was preferred.

- Education: three alternate ILO education definitions were provided: aggregate levels, ISCED-11 and ISCED-97. The aggregate levels (Less than basic, Basic, Intermediate and Advanced, Level not stated) were used, since data for these were provided for all countries and the list is closest to our chosen definitions (basic or less, Intermediate, and advanced).
- Occupations: Both the ISCO-08 and ISCO-88[4] definitions are used, with preference given to the newer definition – ISCO-08 – where both were available. These ILO definitions were mapped to the GTAP occupations and the data aggregated.
- Activities: All 4 activity definitions (ISIC Rev 4, ISIC Rev 3.1, Aggregate, Broad sector). Preference was given to ISIC Rev 4 (22 activities), and when not available ISIC Rev 3.1 (18 activities) was used, or if that was also unavailable the Aggregate (7 activities) or Broad sectors (5 activities) if nothing else was available. The preference structure reflects the higher level of disaggregation of activities. Regardless of the ILO definition of activities used, these activities were further disaggregated using the GTAP wage bill data to obtain data for all 65 GTAP sectors. The assumption used to disaggregate was that wages within the aggregate sector are assumed equal for a given occupation and hence differences in value shares reflect differences in the quantities of labor. This assumption is similar to the one used to construct the wage bill data shares in the GTAP database.

4 Three-stage pipeline

This pipeline builds a globally consistent labor database — population, labor force, employment and unemployment by education, occupation, and economic activity — for use with the GTAP (Global Trade Analysis Project) computable general equilibrium model. It combines a Python data-collection stage with a GEMPACK data-processing stage.

- Stage 1 — Collect (Labor.py): pulls raw indicators from the World Bank API and ILO (ILOSTAT) bulk Excel exports, does light cleanup, and writes each dataset to a GEMPACK HAR file using Harpy.
- Stage 2 — Fill (Labforce.tab, LDocc.tab): reads the raw HAR data at the country level and fills gaps — missing countries, missing education categories, missing occupation/activity breakdowns — using remainder calculations, closest-available-year matching, and regional-average fallback shares. Produces internally consistent, fully populated country-level data.
- Stage 3 — Aggregate (aggregation.tab): maps the filled, detailed, country-level data up to GTAP's official region list and aggregate education categories, producing the final HAR file consumed by the GTAP database-building process.

5 Updating and Improving the Program

A few points to note for anyone who is looking to adapt or improve upon the program to a new version of the GTAP Database:

- The missing population values for 34 regions are hard-coded constants in stage 1 (Labor.py). Because they feed directly into population-based scaling throughout Labforce.tab, it would be necessary to extend the year range when updating to a new GTAP version or different year. Sets.har from the updated GTAP database is also needed.
- The file extradat.har file must be updated with the new header VFM from the latest GTAP database. Alternatively the basedata.har file could be used.

[4] Since all countries included data under at least one of these two preferred definitions, we did not have to use the ILO occupation definition labelled 'skill level'.

- The file specialsets.har file may need to be updated if there is a change in:
    - any of the ILO definitions for education, occupation or activities. This file contains the sets and the mappings for the aggregations and splitting of these sets.
    - This file also contains the long country list and the regional grouping and mapping. This should be updated if there are changes in the country lists. The reasoning behind the regional mapping can be seen in the excel file country_classifictions.xlsx.
- The mapping between the large country list and the GTAP regions by versions are contained in the excel files mapping.xlsx and GTAPmap.xlsx – these would also need t be adapted as GTAP regions or countries change.
- Year encoding: the program assumes the time set's first element corresponds to year 2000. This must stay aligned with Labor.py, which starts every World Bank/ILO pull at year 2000.
- "Best year" selection: whenever a series has data in multiple years but not the target year, the file picks whichever available year is numerically closest to the year of interest (search forward and backward, take the smaller gap), rather than interpolating. This was done because the ILO data often only included one year. If the ILO database improves and more years are collected it could be worth including interpolation of the data in the python program.
- Finally, 2023/2024 population estimates for micro-territories and dependencies not in the World Bank series (e.g. Falkland Islands, Guernsey, Jersey, Taiwan, Western Sahara) are obtained from an internet search and hard coded into the python program. Some of these countries have zero population and hence population gaps are deliberately not filled — any country still showing zero population after the Python stage will propagate zero labor force, employment, etc. downstream. This is intentional since there are small island countries in the country list that have zero population. It is therefore important to ensure that countries with a non-zero population do not have a zero population after the python program (e.g., Taiwan).
- The 60% unemployment-rate guardrail (and its education-category equivalent) is a fixed threshold with no adjustable parameter — this is worth reviewing periodically as new data comes in.
- Preference ordering between alternative occupation and activity classifications are hard-coded in the formula order and not driven by an explicit data-quality metric – this may need to be revisited should revisions be made to the ILO category definitions for education, occupation of activities.